MiniBank (Part B)

- In Account, override the toString() method to return a readable line containing accountNumber, ownerName and balance.
- In Account, override equals(Object o) and hashCode() so that two accounts are equal when their accountNumber values are equal.
- In Customer, add a public static nested class named Address with String fields line, city and pincode plus getters; add an Address field to Customer and a getAddress() method.
- In Customer, add a clone() method (implement Cloneable) that returns a copy of the customer.
- In main, print accounts using toString(), compare two Account objects with equals(), and use instanceof to check an object's type.

Source Code (Insert new code inside old MiniBank Practical 2 Files):

Account.java:

```
// --------- Practical - 3 ----------------
public String toString() {
    return "Account{"
            + "accountNumber='" + accountNumber + '\''
            + ", ownerName='" + ownerName + '\''
            + ", balance=" + balance
            + '}';
}


   public boolean equals(Object o) {
       if (this == o) {
           return true;
       }
       if (!(o instanceof Account)) {
           return false;
       }

       Account account = (Account) o;
       return accountNumber.equals(account.accountNumber);
   }

   public int hashCode() {
       return Objects.hash(accountNumber);
   }

}
```

Customer.java:

```
// ---------- Practical - 3 ----------------

public void setAddress(Address address) {
    this.address = address;
}

public Address getAddress() { return address; }
You, 18 minutes ago | author (You)
public static class Address {
    private String line;
    private String city;
    private String pincode;

    public Address(String line, String city, String pincode) {
        this.line = line;
        this.city = city;
        this.pincode = pincode;
    }
    public String getFullAddress(){ return line+", "+city+", "+pincode; }
    public String getLine() { return line; }
    public String getCity() { return city; }
    public String getPincode() { return pincode; }
}

public Customer clone() {
    try {
        return (Customer) super.clone();
    } catch (CloneNotSupportedException e) {
        throw new AssertionError(e);
    }
}
```

MiniBank.java:

```
// ==========================================
// PRACTICAL 3 TESTING
// ==========================================

Account account1 = new Account(ownerName: "Yash", openingBalance: 5000);
Account account2 = new Account(ownerName: "Rahul", openingBalance: 10000);

// toString()
System.out.println(x: "\n===== toString() TEST =====");
System.out.println(account1);
System.out.println(account2);

// equals()
System.out.println(x: "\n===== equals() TEST =====");
System.out.println("Are account1 and account2 equal? " + account1.equals(account2));
System.out.println("Are account1 and account1 equal? " + account1.equals(account1));

// instanceof
System.out.println(x: "\n===== instanceof TEST =====");
Object obj = account1;      You, 15 minutes ago • Uncommitted changes
if (obj instanceof Account) {
    System.out.println(x: "obj is an Account object.");
} else {
    System.out.println(x: "obj is not an Account object.");
}

// Customer and Address test
System.out.println(x: "\n===== CUSTOMER TEST =====");
Customer customer = new Customer(name: "Yash", email: "yash@example.com", mobile: "9876543210");
Customer.Address address = new Customer.Address(line: "CSPIT Campus", city: "Anand", pincode: "388001");
customer.setAddress(address);
System.out.println("Customer ID : " + customer.getCustomerId());
System.out.println("Name        : " + customer.getName());
System.out.println("Email       : " + customer.getEmail());
System.out.println("Mobile      : " + customer.getMobile());
System.out.println("Address     : " + customer.getAddress().getFullAddress());

// Clone test
Customer clonedCustomer = customer.clone();
System.out.println(x: "\n===== CLONE TEST =====");
System.out.println("Original Customer ID : " + customer.getCustomerId());
System.out.println("Cloned Customer ID   : " + clonedCustomer.getCustomerId());
System.out.println("Is clone a different object? " + (customer != clonedCustomer));
// Put test accounts into account array
accounts[accountCount++] = account1;
accounts[accountCount++] = account2;
```

MiniBank Output:

```
PS E:\SEM-3 CSPIT\OOP\Practicals> javac .\MiniBank\*.java
PS E:\SEM-3 CSPIT\OOP\Practicals> java MiniBank.MiniBank
====================================
BankInfo[name=MiniBank, branch=Anand Branch]
====================================

===== toString() TEST =====
Account{accountNumber='AC0001', ownerName='Jigar', balance=5000}
Account{accountNumber='AC0002', ownerName='Jigar Kotecha', balance=10000}

===== equals() TEST =====
Are account1 and account2 equal? false
Are account1 and account2 equal? true

===== instanceof TEST =====
obj is an Account object.

===== CUSTOMER TEST =====
Customer ID : CUST101
Name        : Jigar Kotecha
Email       : jigar@example.com
Mobile      : 9876543210
Address     : CSPIT Campus, Anand, 388001

===== CLONE TEST =====
Original Customer ID : CUST101
Cloned Customer ID   : CUST101
Is clone a different object? true

===== MINI BANK MENU =====
1. Open Account
2. Deposit
3. Withdraw
4. Transfer
5. Show Account Details
6. Exit
Enter Choice : 5

===== ACCOUNT DETAILS =====
Account Number : AC0001
Owner Name     : Jigar
Balance        : 5000
Active         : true
----------------------------
Account Number : AC0002
Owner Name     : Jigar Kotecha
Balance        : 10000
Active         : true
PS E:\SEM-3 CSPIT\OOP\Practicals>
```

GitHub Repository (MiniBank):

https://github.com/d26it118-png/OOP_PRACTICALS.git